Decision Tree

Problem:

Suppose you are given following data that determines whether a person goes out and play based on following weather conditions:
Outlook, Humidity and Wind

Using this data, construct a decision tree that would decide based on features (Outlook, Humidity and Wind) whether a person would play or not.

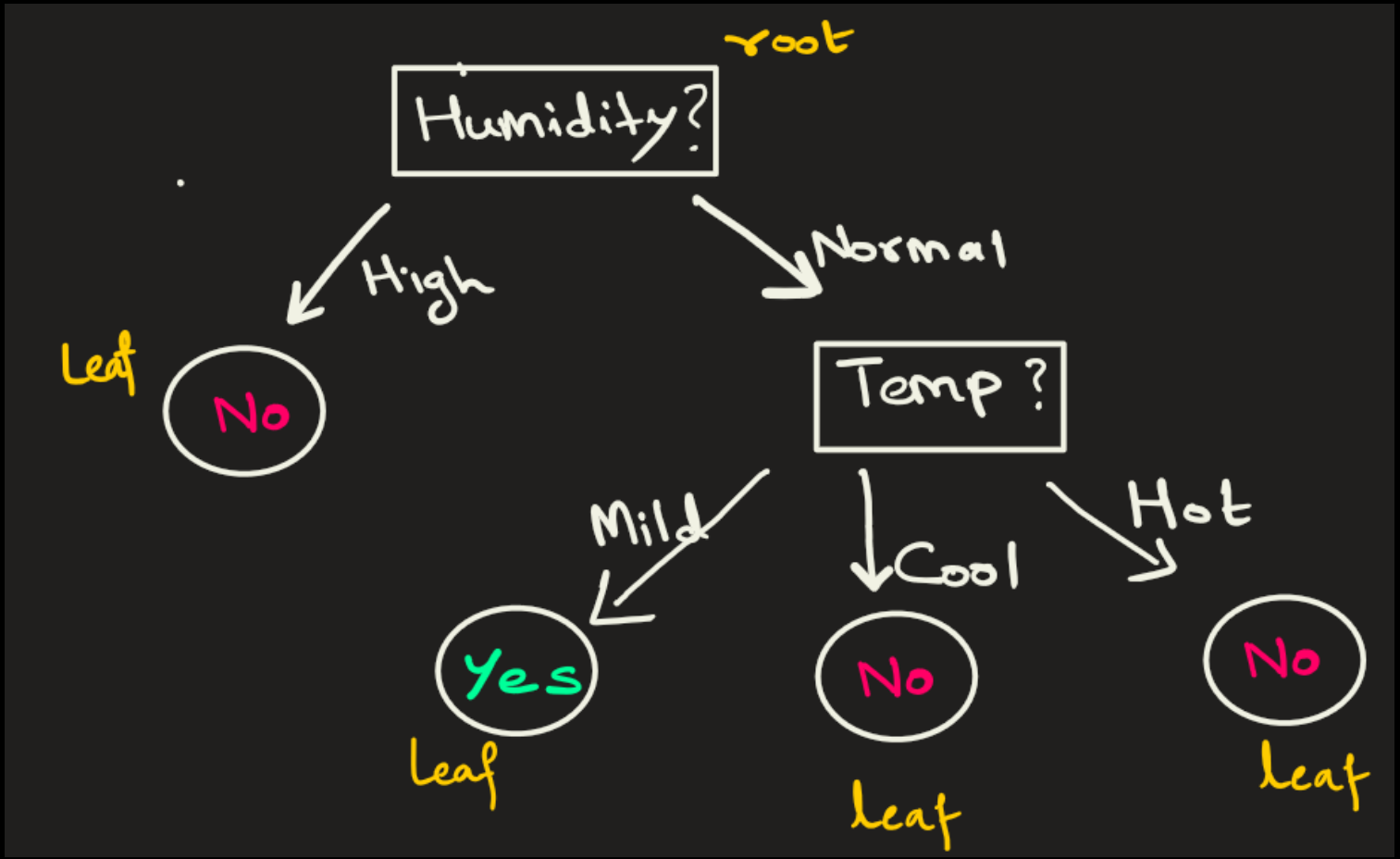
ID	Outlook	Humidity	Wind	Play
1	Sunny	High	Weak	No
2	Sunny	High	Strong	No
3	Overcast	High	Weak	Yes
4	Rain	High	Weak	Yes
5	Rain	Normal	Weak	Yes
6	Rain	Normal	Strong	No
7	Overcast	Normal	Strong	Yes
8	Sunny	Normal	Weak	Yes

Decision Tree

A **Decision Tree** is a supervised machine learning algorithm used for **classification and regression**.

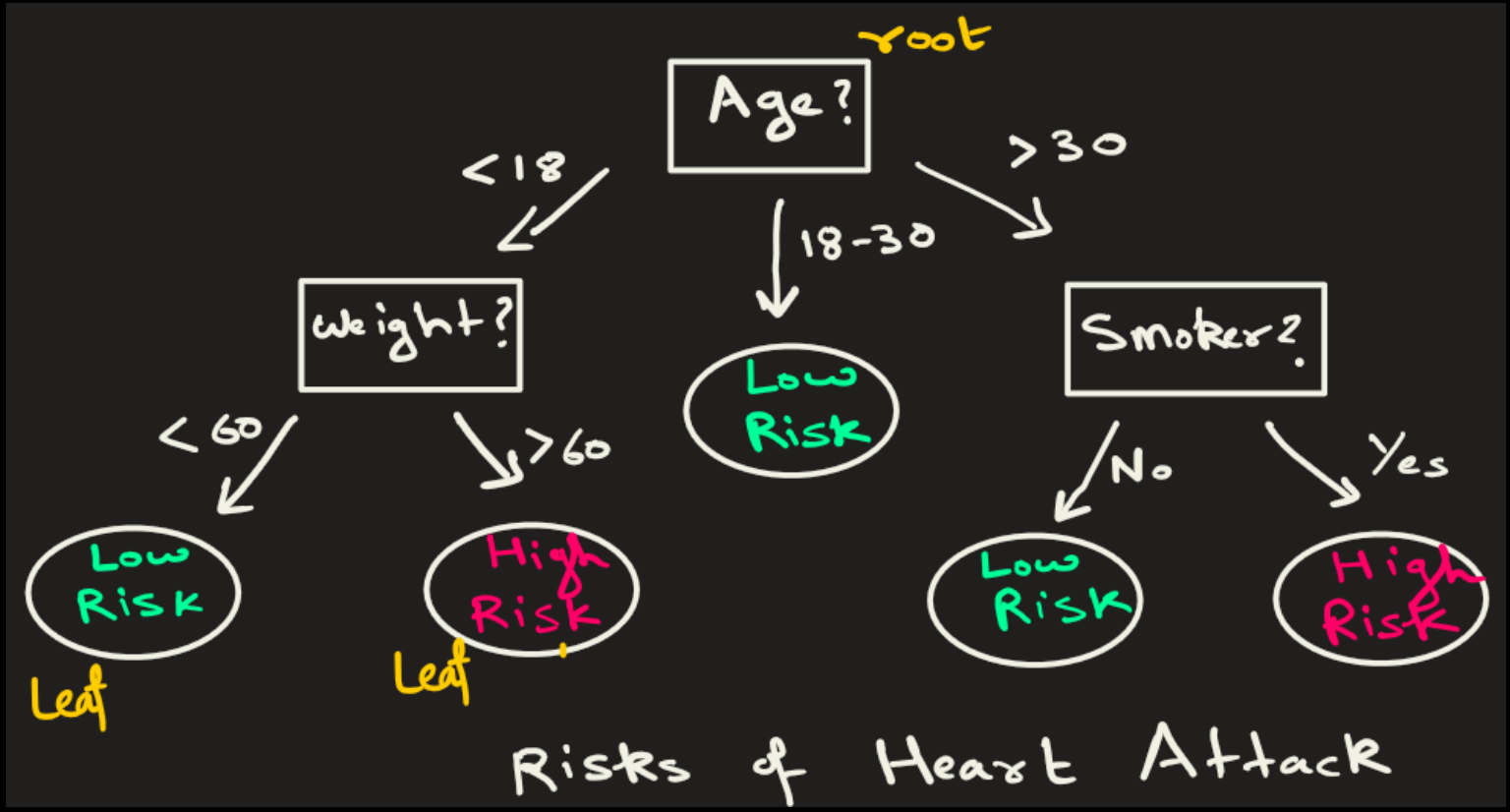
Decision Tree: Idea

Suppose you have to decide if you are going to play outside.
You make your decision based on 2 features only: Humidity and Temperature
Humidity can be high or normal.
Temperature can be cool, mild or hot



Decision Tree

Following is another example of decision :
Features are age, weight and smoker.
Root node starts with age, then in left node it decides based on wt.
Right node decides based on smoker.
The response variable is High/Low risk of heart attack



Decision Tree: Algorithm

How is a decision tree constructed ? It is constructed from training data.

1) Choose the Best Attribute:
The algorithm evaluates all features using a metric such as **Gini impurity**, **entropy**, or **information gain**.
The feature that **best separates the classes** becomes the **root node** (the first decision).

2) Split the Dataset:
The dataset is divided into groups based on the selected feature.

- For numerical features → a **threshold** is used (e.g., in Iris dataset, petal_length <= **2.5**)
- For categorical features → the data is split by categories (e.g., Temperature = Hot, Mild, Cool)

3) Repeat the Process:
For each new subset of data, the same steps are applied again:
Choose the best feature → Split the data → Create new branches

This continues until a **stopping rule** is reached, such as:

- All samples in a node belong to the same class
- No features left to split
- Maximum depth reached
- Minimum samples per node reached

Entropy: It measures the randomness in data

$$H(S) = -\sum_{i=1}^{k} p_i \log_2(p_i)$$

Symbol	Meaning
S	The dataset (Entire or subset)
k	Number of class labels (e.g., Banana/Not-Banana → k=2)
p_i	The probability of class i in the dataset S

Example: Suppose we have 2 class: + and -. Then the formula becomes,

$i = 1$ +

$i = 2$ −

$$\boxed{\begin{matrix} + & + & + & + & + \\ - & - & - & - & - \end{matrix}}$$

$$H(S) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -[p_1 \log p_1 + p_2 \log p_2]$$

Below we see that as randomness is lowering, entropy H(S) is getting lower.
For **Binary class**, the entropy ranges from 0 to +1.

$i=1$ +
$i=2$ −

+ + + + +
− − − − −

$$H(S) = -\sum_{i=1}^{2} p_i \log p_i$$
$$= -[p_1 \log p_1 + p_2 \log p_2]$$
$$= -\left[\frac{5}{10}\log\frac{5}{10} + \frac{5}{10}\log\frac{5}{10}\right]$$
$$= +1 \text{ (Maximum)}$$

+ + + + +
+ + + + −

$$H(S) = -\sum_{i=1}^{2} p_i \log p_i$$
$$= -[p_1 \log p_1 + p_2 \log p_2]$$
$$= -\left[\frac{9}{10}\log\frac{9}{10} + \frac{1}{10}\log\frac{1}{10}\right]$$
$$= 0.469$$

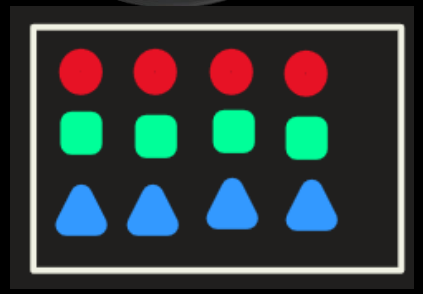
+ + + + +
+ + − − −

$$H(S) = -\sum_{i=1}^{2} p_i \log p_i$$
$$= -[p_1 \log p_1 + p_2 \log p_2]$$
$$= -\left[\frac{7}{10}\log\frac{7}{10} + \frac{3}{10}\log\frac{3}{10}\right]$$
$$= +0.8812$$

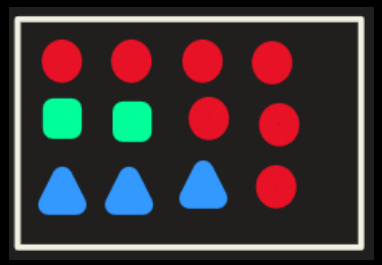
+ + + + +
+ + + + +

$$H(S) = -\sum_{i=1}^{2} p_i \log p_i$$
$$= -[p_1 \log p_1 + p_2 \log p_2]$$
$$= -\left[\frac{10}{10}\log\frac{10}{10} + \frac{0}{10}\log\frac{0}{10}\right]$$
$$= 0$$

Example: Below we have 3 classes
For tertiary class, the entropy ranges from 0 to +1.585



$p_1 \rightarrow$ prob. of red
$p_2 \rightarrow$ prob. of green
$p_3 \rightarrow$ prob. of blue



$$H(S) = -\sum_{i=1}^{3} p_i \log p_i$$

$$= -[p_1 \log p_1 + p_2 \log p_2 + p_3 \log p_3]$$

$$= -\left[\frac{4}{12}\log\left(\frac{4}{12}\right) + \frac{4}{12}\log\frac{4}{12} + \frac{4}{12}\log\left(\frac{4}{12}\right)\right]$$

$$= +1.585$$

$$H(S) = -\sum_{i=1}^{3} p_i \log p_i$$

$$= -[p_1 \log p_1 + p_2 \log p_2 + p_3 \log p_3]$$

$$= -\left[\frac{7}{12}\log\left(\frac{7}{12}\right) + \frac{2}{12}\log\left(\frac{2}{12}\right) + \frac{3}{12}\log\left(\frac{3}{12}\right)\right]$$

$$= +1.384$$

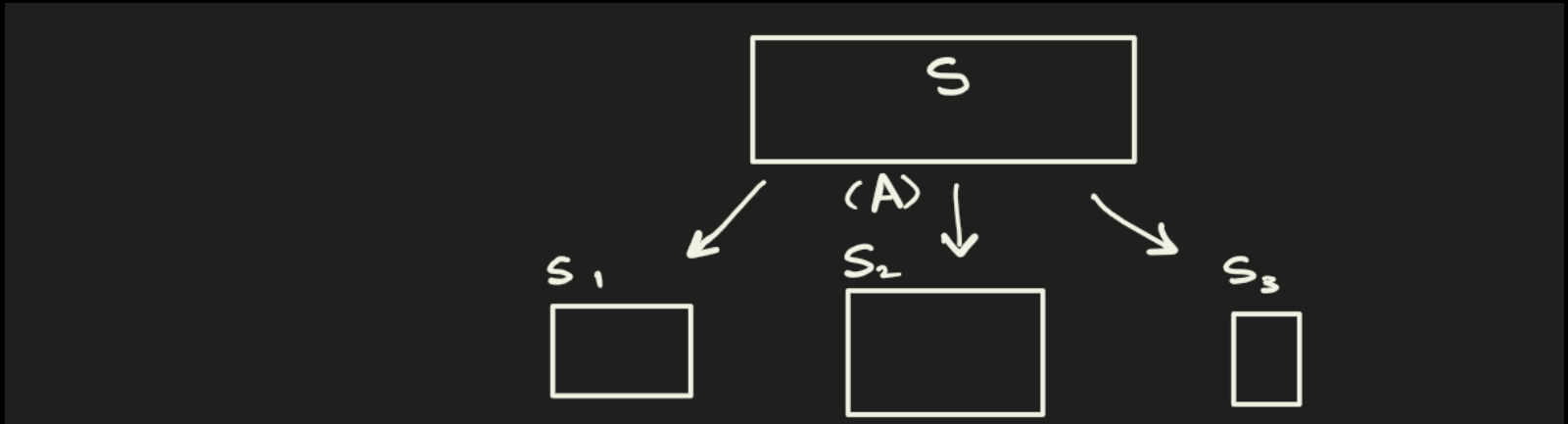
Information gain

Information Gain = **Entropy before split – Entropy after split**
= **Entropy of Parent – Weighted entropy of child nodes**

$$IG(S,A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

Symbol	Meaning
IG(S,A)	Information Gain of attribute **A**
H(S)	Entropy of the original dataset
Values(A)	All possible values of attribute **A**
S_v	Subset of data where attribute A = v
H(S_v)	Entropy of subset S_v

Example: Suppose that S is split into 3 child nodes. Then formula for IG is



$$I.G(S,A) = H(S) - \left[\frac{|S_1|}{|S|} H(S_1) + \frac{|S_2|}{|S|} H(S_2) + \frac{|S_3|}{|S|} H(S_3) \right]$$

Information gain

Suppose you are given following training data. Here the features are weight, shape and sweetness. Using this data, construct a decision tree that would decide based on features whether a fruit is a banana or not.

At any node, how do you determine which feature to use when splitting?

weight	shape	sweet	is_banana
0.1	round	y	Not banana
0.2	round	y	Not banana
0.2	round	n	Not banana
0.2	round	y	Not banana
0.4	elongated	y	Banana
0.6	elongated	n	Banana
0.1	elongated	y	Banana
0.6	round	n	Not banana
0.4	elongated	n	Banana
0.5	elongated	n	Banana

Here **target** variable(aka, **response** or **label**) is **is_banana**

The **shape** is best feature that determines the type of fruit. So, this feature has most **information gain** among all. So, start by asking if the fruit **shape** is elongated or round, we can quickly determine the **type** of fruit.
This would be good candidate for splitting.

Feature **sweet** provides no useful information on the type of fruit.

Information gain

weight	shape	sweet	is_banana
0.1	round	y	Not banana
0.2	round	y	Not banana
0.2	round	n	Not banana
0.2	round	y	Not banana
0.4	elongated	y	Banana
0.6	elongated	n	Banana
0.1	elongated	y	Banana
0.6	round	n	Not banana
0.4	elongated	n	Banana
0.5	elongated	n	Banana

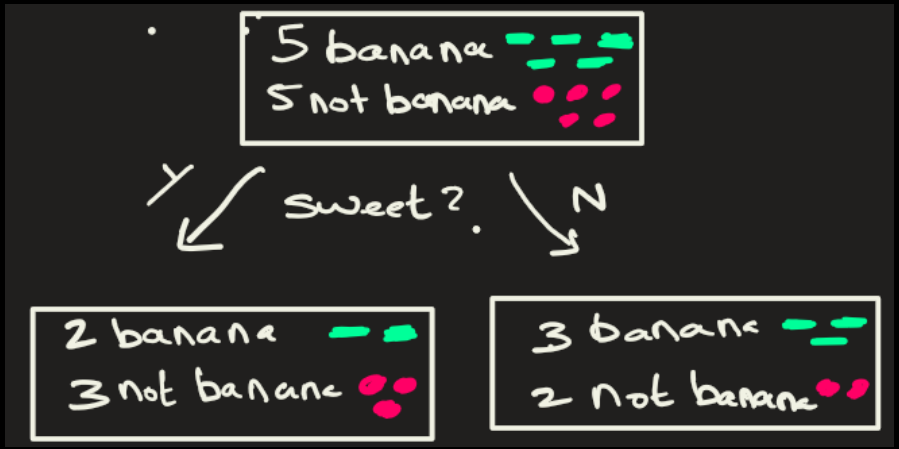
$$IG = H(\text{Parent}) - \left[\frac{|\text{L-child}|}{|\text{parent}|} \cdot H(\text{L-child}) + \frac{|\text{R-child}|}{|\text{parent}|} \cdot H(\text{R-child}) \right]$$

$$H(\text{Parent}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log p_2\right] \quad (p_1 \to \text{prob. of banana})$$

$$= -\left[\left(\frac{5}{10}\right)\log\left(\frac{5}{10}\right) + \left(\frac{5}{10}\right)\cdot\log\left(\frac{5}{10}\right)\right] \quad (p_2 \to \text{prob. of } \underline{\text{not}} \text{ banana})$$

$$= +1$$



$$H(\text{L-child}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log_2 p_2\right]$$

$$= -\left[\frac{2}{5}\log\left(\frac{2}{5}\right) + \frac{3}{5}\log\left(\frac{3}{5}\right)\right]$$

$$= 0.9709$$

$$H(\text{R-child}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log p_2\right]$$

$$= -\left[\frac{3}{5}\log\left(\frac{3}{5}\right) + \frac{2}{5}\log\left(\frac{2}{5}\right)\right]$$

$$= 0.9709$$

Less Information Gain from feature sweet
Final split is more impure.

$$IG = 1 - \left[\frac{5}{10} \times 0.9709 + \frac{5}{10} \times 0.9709\right] = 0.029$$

Information gain

weight	shape	sweet	is_banana
0.1	round	y	Not banana
0.2	round	y	Not banana
0.2	round	n	Not banana
0.2	round	y	Not banana
0.4	elongated	y	Banana
0.6	elongated	n	Banana
0.1	elongated	y	Banana
0.6	round	n	Not banana
0.4	elongated	n	Banana
0.5	elongated	n	Banana

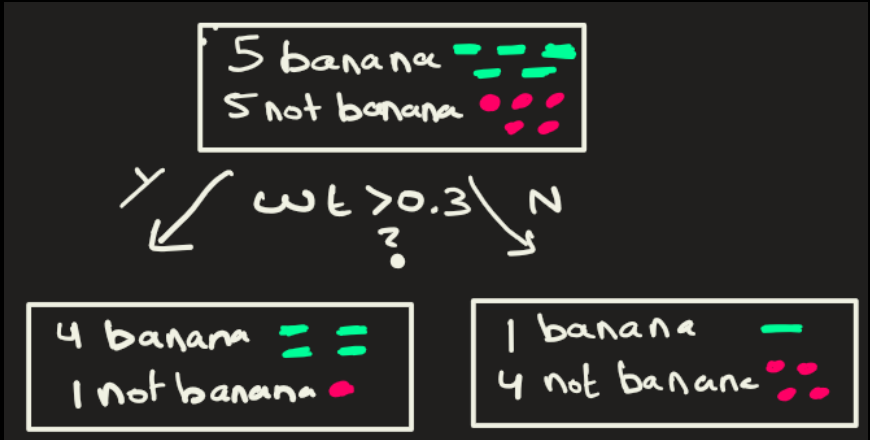
$$IG = H(\text{Parent}) - \left[\frac{|\text{L-child}|}{|\text{parent}|} \cdot H(\text{L-child}) + \frac{|\text{R-child}|}{|\text{parent}|} \cdot H(\text{R-child}) \right]$$

$$H(\text{Parent}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log p_2\right] \quad (p_1 \to \text{prob. of banana})$$

$$= -\left[\left(\frac{5}{10}\right)\log\left(\frac{5}{10}\right) + \left(\frac{5}{10}\right)\log\left(\frac{5}{10}\right)\right] \quad (p_2 \to \text{prob. of } \underline{\text{not}} \text{ banana})$$

$$= +1$$



$$H(\text{L-child}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log p_2\right]$$

$$= -\left[\frac{4}{5}\log\left(\frac{4}{5}\right) + \frac{1}{5}\log\left(\frac{1}{5}\right)\right]$$

$$= 0.7219$$

$$H(\text{R-child}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log p_2\right]$$

$$= -\left[\frac{1}{5}\log\left(\frac{1}{5}\right) + \frac{4}{5}\log\left(\frac{4}{5}\right)\right]$$

$$= 0.7219$$

Information Gain increases from feature weight.
Final split is less Impure

$$I.G. = 1 - \left[\frac{5}{10} \times 0.7219 + \frac{5}{10} \times 0.7219\right] = 0.2781$$

Information gain

weight	shape	sweet	is_banana
0.1	round	y	Not banana
0.2	round	y	Not banana
0.2	round	n	Not banana
0.2	round	y	Not banana
0.4	elongated	y	Banana
0.6	elongated	n	Banana
0.1	elongated	y	Banana
0.6	round	n	Not banana
0.4	elongated	n	Banana
0.5	elongated	n	Banana

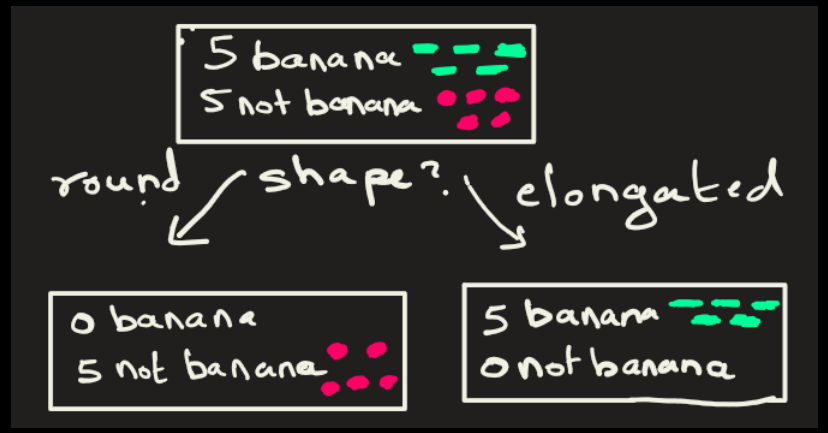
$$IG = H(\text{Parent}) - \left[\frac{|\text{L-child}|}{|\text{parent}|} \cdot H(\text{L-child}) + \frac{|\text{R-child}|}{|\text{parent}|} \cdot H(\text{R-child}) \right]$$

$$H(\text{Parent}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log p_2\right] \quad (p_1 \to \text{prob. of banana})$$

$$= -\left[\left(\frac{5}{10}\right)\log\left(\frac{5}{10}\right) + \left(\frac{5}{10}\right)\log\left(\frac{5}{10}\right)\right] \quad (p_2 \to \text{prob. of } \underline{\text{not}} \text{ banana})$$

$$= +1$$



$$H(\text{L-child}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log p_2\right]$$

$$= -\left[0 + \frac{5}{5}\log\left(\frac{5}{5}\right)\right]$$

$$= \underline{\underline{0}}$$

$$H(\text{R-child}) = -\sum_{i=1}^{2} p_i \log p_i$$

$$= -\left[p_1 \log p_1 + p_2 \log p_2\right]$$

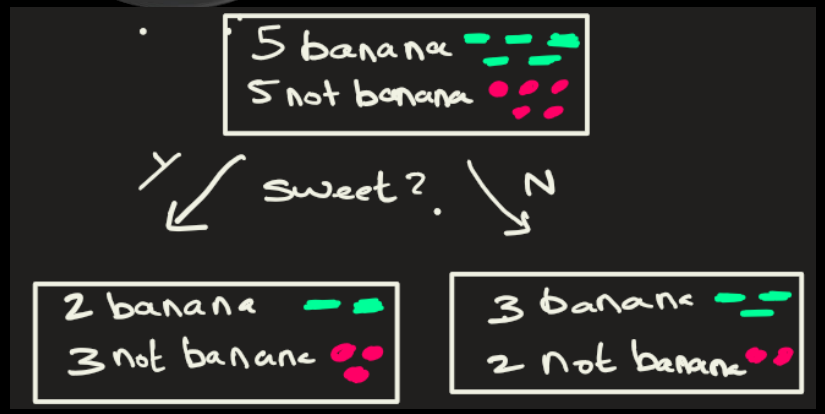
$$= -\left[\frac{5}{5}\log\left(\frac{5}{5}\right) + 0\right]$$

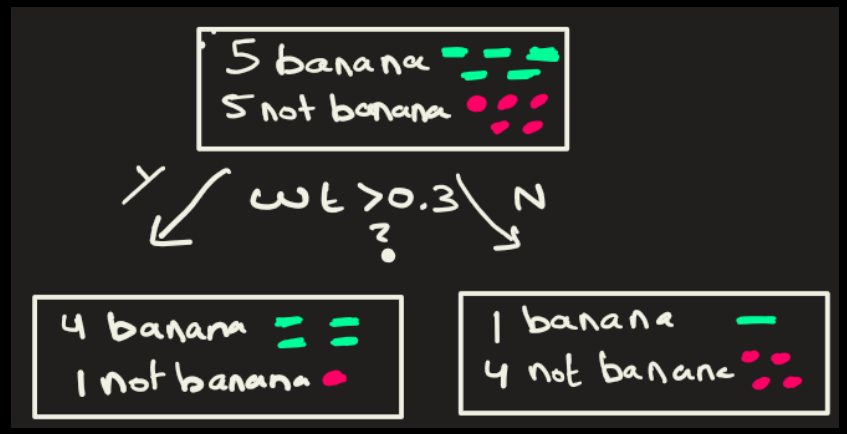
$$= \underline{\underline{0}}$$

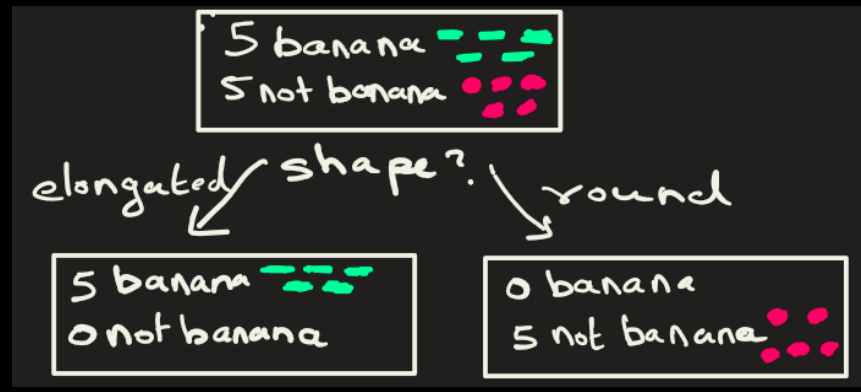
$$IG = 1 - [0 + 0] = \underline{\underline{1}}$$

Maximum information gain from shape
Final split is **pure**.

Information gain





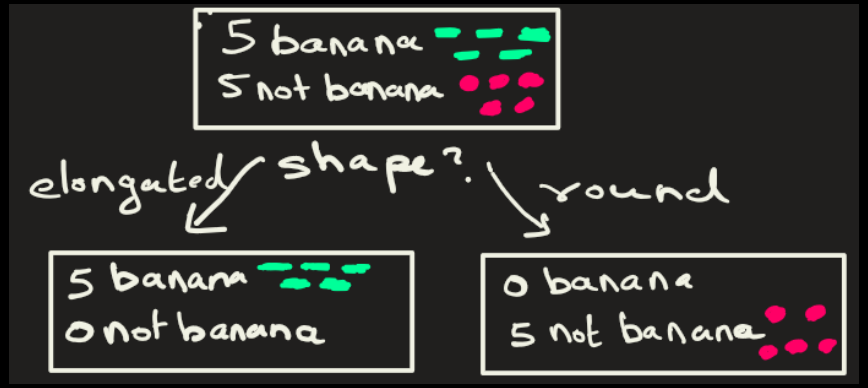


IG(Parent, ‘sweet’) = 0.029

IG(Parent, ‘weight’) = 0.2781

IG(Parent, ‘shape’) = 1

So, we choose shape as the attribute to split the parent as it gives maximum IG. The split is pure.



Decision Tree

$$H(S) = -\sum_{i=1}^{k} p_i \log_2(p_i)$$

$$IG(S,A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

Example: Create a decision tree from following data

ID	Outlook	Humidity	Wind	Play
1	Sunny	High	Weak	No
2	Sunny	High	Strong	No
3	Overcast	High	Weak	Yes
4	Rain	High	Weak	Yes
5	Rain	Normal	Weak	Yes
6	Rain	Normal	Strong	No
7	Overcast	Normal	Strong	Yes
8	Sunny	Normal	Weak	Yes

Decision Tree

ID	Outlook	Humidity	Wind	Play
1	Sunny	High	Weak	No
2	Sunny	High	Strong	No
3	Overcast	High	Weak	Yes
4	Rain	High	Weak	Yes
5	Rain	Normal	Weak	Yes
6	Rain	Normal	Strong	No
7	Overcast	Normal	Strong	Yes
8	Sunny	Normal	Weak	Yes

STEP 1 — Calculate Entropy of the Full Dataset

$$H = -p_{yes}\log_2(p_{yes}) - p_{no}\log_2(p_{no})$$

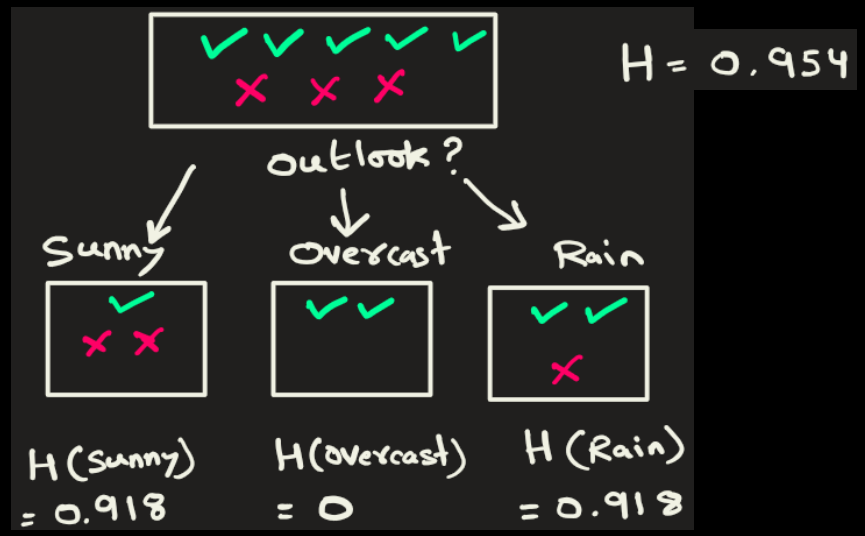
$$H = -\frac{5}{8}\log_2(0.625) - \frac{3}{8}\log_2(0.375) = 0.954$$

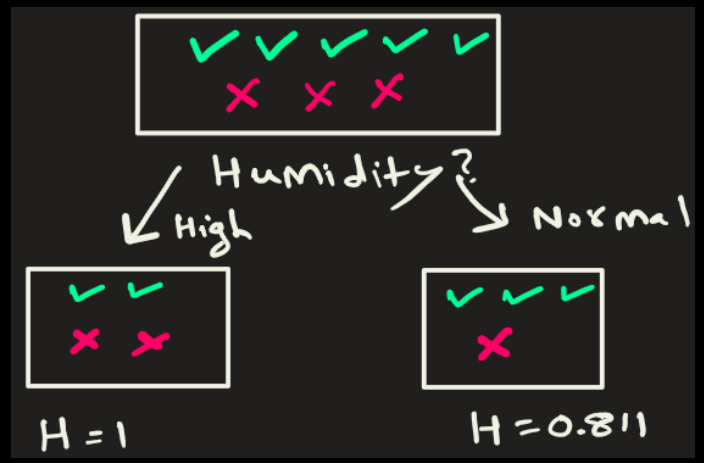
Total entropy of parent = 0.954

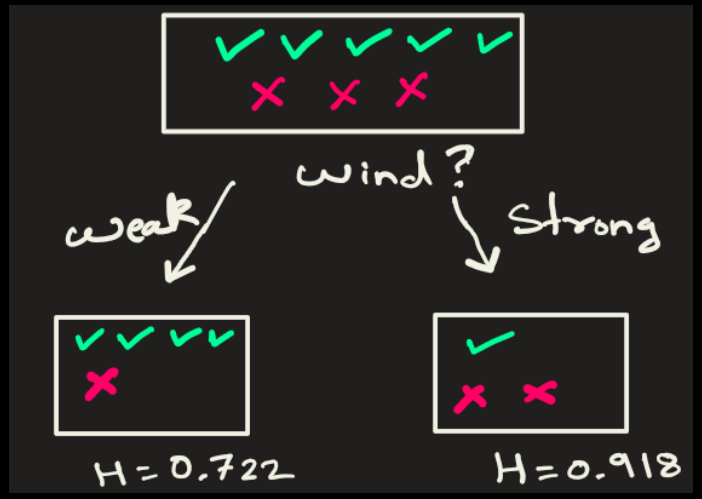
Decision Tree

$$IG(S,A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

STEP 2 — Compute Information Gain for each feature: outlook, humidity and wind







Weighted Entropy

$$H_{outlook} = \frac{3}{8}(0.918) + \frac{2}{8}(0) + \frac{3}{8}(0.918)$$
$$= 0.689$$

Weighted Entropy

$$H_{wind} = \frac{4}{8}(1) + \frac{4}{8}(0.811)$$
$$= 0.905$$

Weighted Entropy

$$H_{wind} = \frac{5}{8}(0.722) + \frac{3}{8}(0.918)$$
$$= 0.799$$

$$IG = 0.954 - 0.689 = 0.265$$

$$IG = 0.954 - 0.905 = 0.049$$

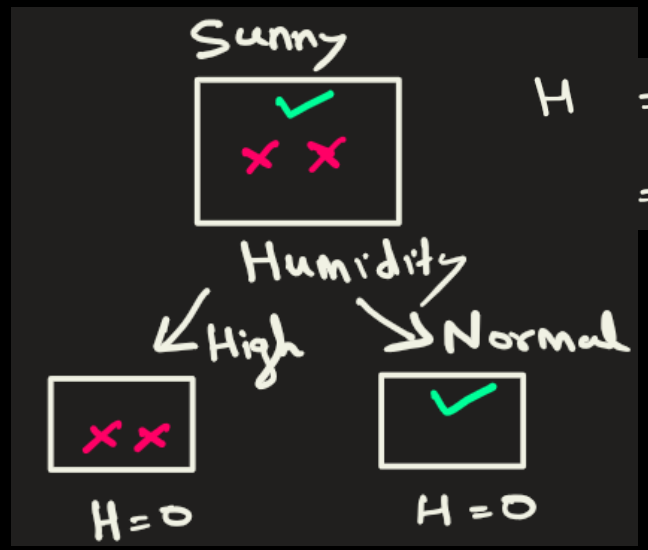
$$IG = 0.954 - 0.799 = 0.155$$

Highest IG = Outlook → becomes Root Node

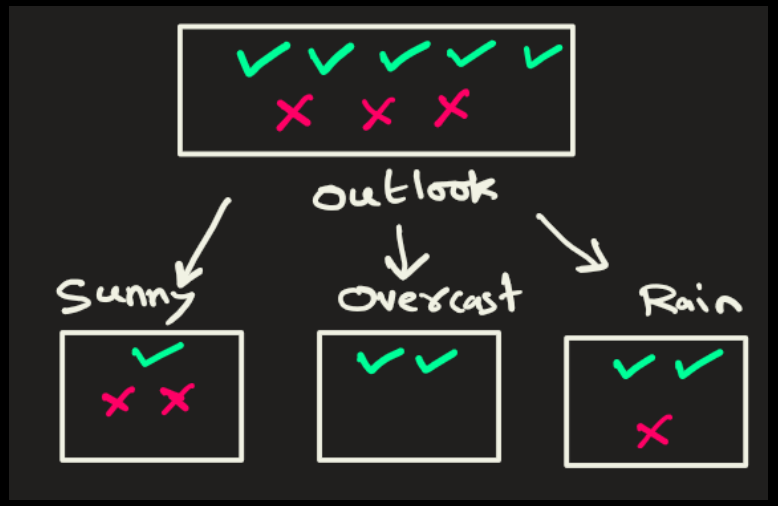
Decision Tree

$$IG(S,A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

STEP 3 — a) Split the dataset on branch Outlook → Sunny



$$H = -\left[\frac{1}{3}\log\left(\frac{1}{3}\right) + \frac{2}{3}\log\left(\frac{2}{3}\right)\right]$$
$$= 0.9235$$



Weighted Entropy

$H_{Humidity} = 0$

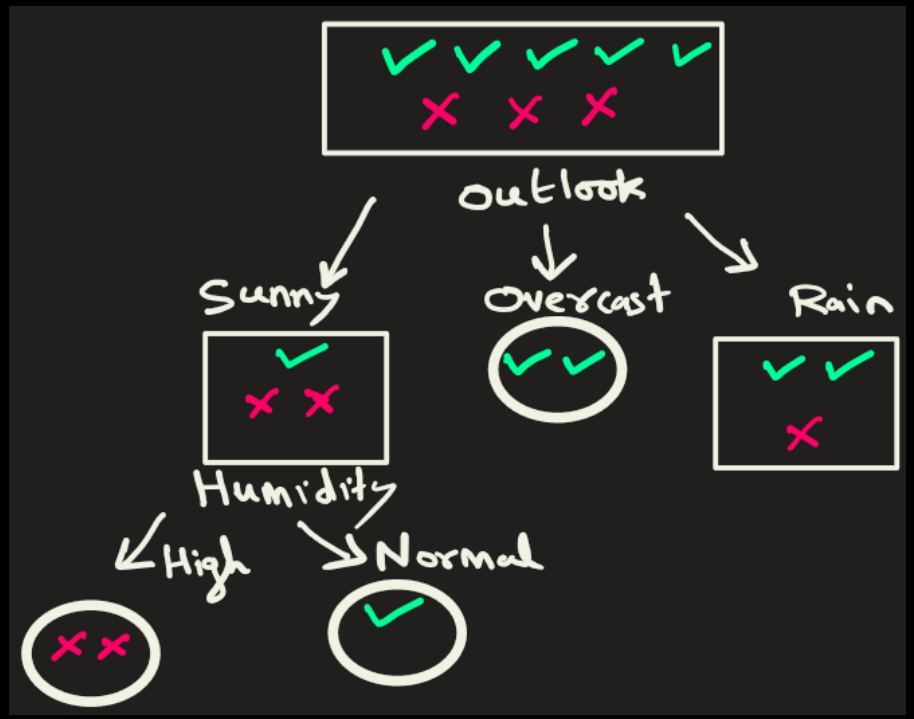
$IG = 0.9235 - 0 = 0.9235$

(This is maximum)

Decision Tree

$$IG(S,A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

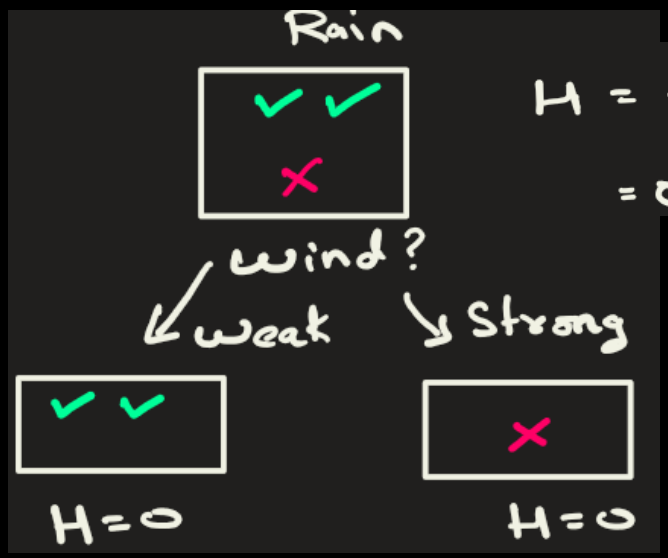
STEP 3 — b) Split the dataset on branch Outlook → Overcast:
Overcast is pure. No split needed here.



Decision Tree

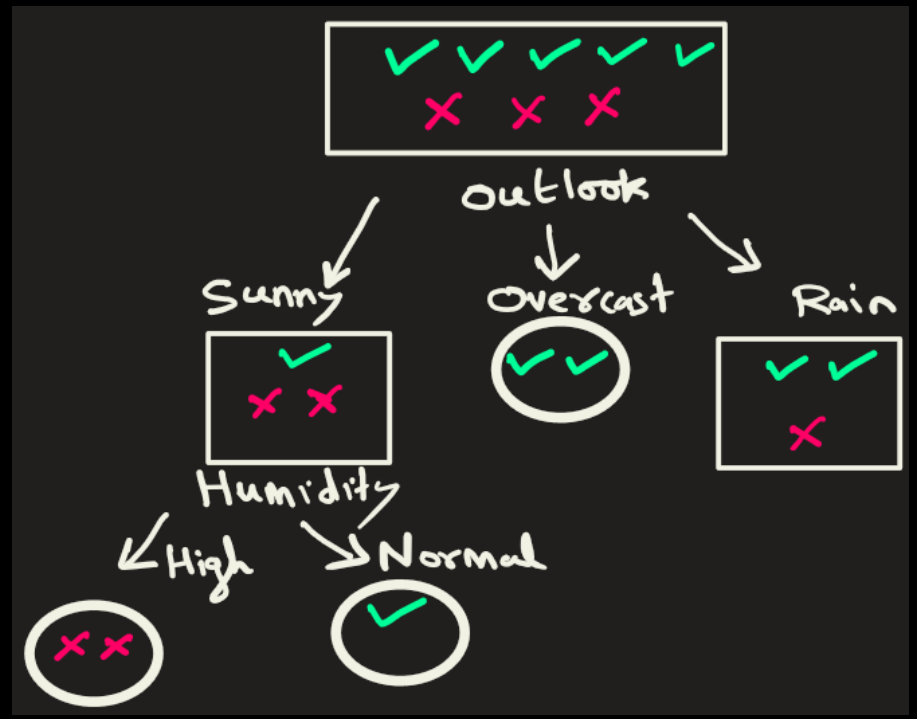
$$IG(S,A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

STEP 3 — c) Split the dataset on branch Outlook → Rain:



$$H = -\left[\frac{2}{3}\log\left(\frac{2}{3}\right) + \frac{1}{3}\log\left(\frac{1}{3}\right)\right]$$

$$= 0.9235$$



Weighted Entropy

$H_{wind} = 0$

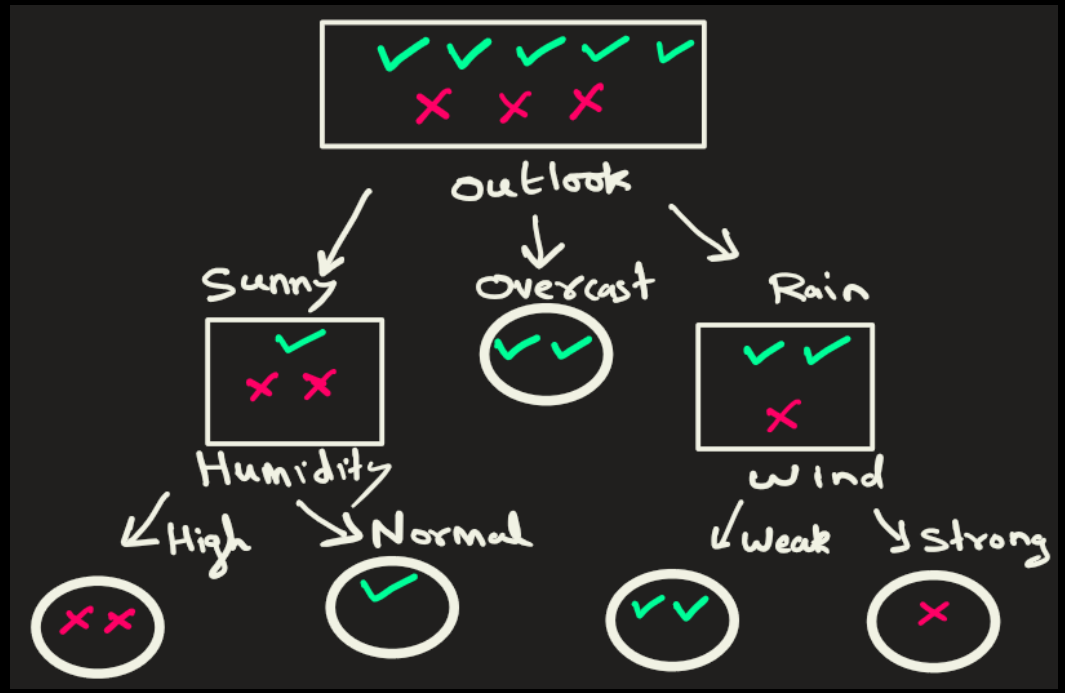
$IG = 0.9235 - 0 = 0.9235$

(IG is maximum)

Decision Tree

$$IG(S,A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)$$

STEP 4 — All splits are pure. No further split required.



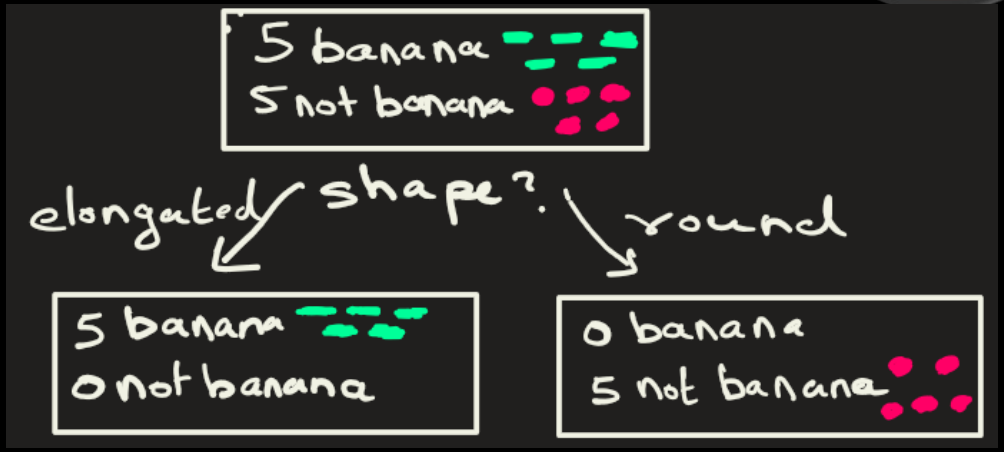
Decision Tree: Advantages

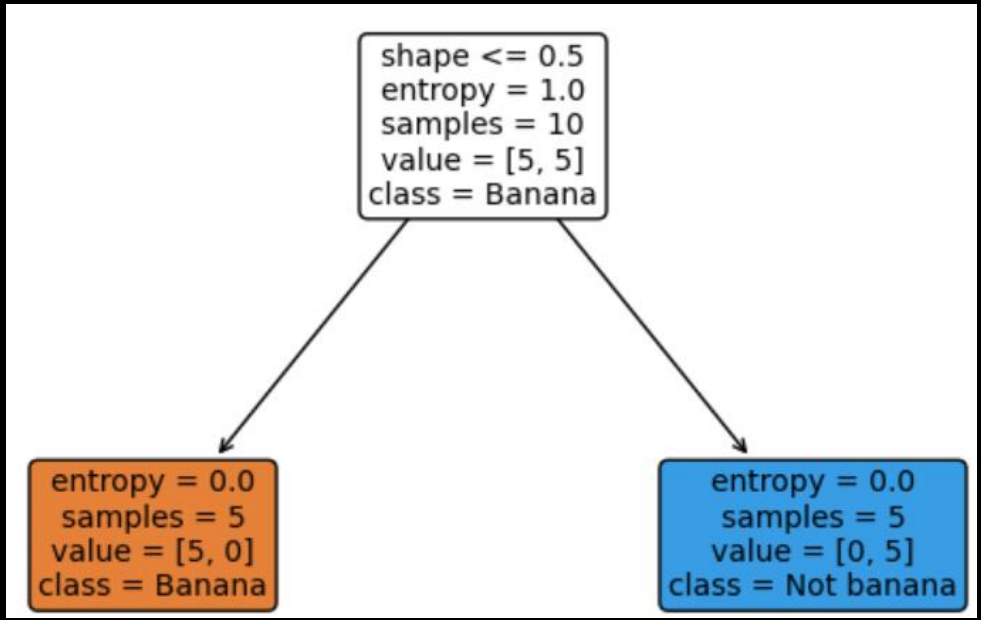
- Easy to understand, visualize, and explain: They use if-else logic
- No need for feature scaling
- Handles numerical & categorical data
- Works well even with missing values and even if outliers exist
- Fast training and prediction

Extra

Decision Tree: Code

```python
from sklearn.tree            import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing    import LabelEncoder
import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_excel("data_fruit_type.xlsx")
X = df[["weight", "shape", "sweet"]]
y = df["is_banana"]
le_y = LabelEncoder()
y_encoded = le_y.fit_transform(y)

X_encoded = X.copy()
le_shape = LabelEncoder()
le_sweet = LabelEncoder()
X_encoded["shape"] = le_shape.fit_transform(X["shape"])
X_encoded["sweet"] = le_sweet.fit_transform(X["sweet"])

X_train, y_train = X_encoded, y_encoded
model = DecisionTreeClassifier(criterion="entropy")
model.fit(X_train, y_train)

plot_tree( model,filled=True, rounded=True, feature_names=X.columns,
                class_names=le_y.classes_, fontsize=10)
plt.show()
```





Steps to calculate IG on Heart disease:

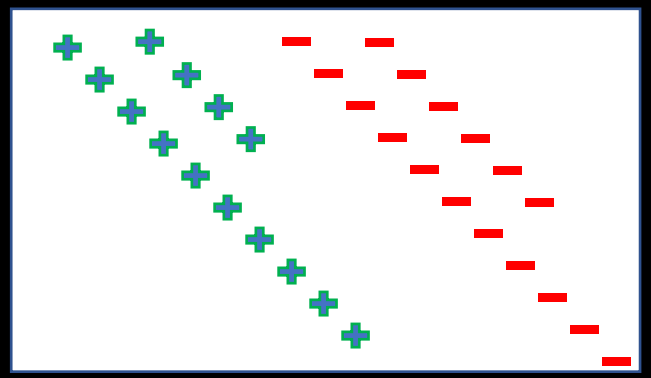
This is an example related to predicting heart disease in patients. At this node, there are 3 features where split could be made:

1) **Age Group** Split: Older individuals are more prone to heart disease.
2) **Smoking Status** Split: Smokers have a higher risk.
3) **Exercise Habit** Split: Lack of exercise increases risk significantly.

Let's determine on which feature the split should be made.

ID	Age Group	Smoking Status	Exercise Habit	Heart Disease (Label)
1	Younger (<40)	Smoker	Regular	Positive
2	Younger (<40)	Smoker	Regular	Positive
3	Younger (<40)	Smoker	Regular	Positive
4	Younger (<40)	Smoker	No Exercise	Positive
5	Younger (<40)	Smoker	No Exercise	Positive
6	Younger (<40)	Non-Smoker	Regular	Positive
7	Younger (<40)	Non-Smoker	Regular	Positive
8	Younger (<40)	Non-Smoker	Regular	Positive
9	Younger (<40)	Non-Smoker	No Exercise	Positive
10	Younger (<40)	Non-Smoker	No Exercise	Positive
11	Younger (<40)	Smoker	No Exercise	Negative
12	Younger (<40)	Smoker	No Exercise	Negative
13	Younger (<40)	Non-Smoker	No Exercise	Negative
14	Younger (<40)	Non-Smoker	No Exercise	Negative
15	Younger (<40)	Non-Smoker	No Exercise	Negative
16	Older (≥40)	Smoker	No Exercise	Positive
17	Older (≥40)	Smoker	No Exercise	Positive
18	Older (≥40)	Non-Smoker	Regular	Positive
19	Older (≥40)	Non-Smoker	Regular	Positive
20	Older (≥40)	Smoker	No Exercise	Negative
21	Older (≥40)	Smoker	No Exercise	Negative
22	Older (≥40)	Smoker	No Exercise	Negative
23	Older (≥40)	Smoker	No Exercise	Negative
24	Older (≥40)	Smoker	No Exercise	Negative
25	Older (≥40)	Smoker	No Exercise	Negative
26	Older (≥40)	Smoker	No Exercise	Negative
27	Older (≥40)	Non-Smoker	No Exercise	Negative
28	Older (≥40)	Non-Smoker	No Exercise	Negative
29	Older (≥40)	Non-Smoker	No Exercise	Negative
30	Older (≥40)	Non-Smoker	No Exercise	Negative

Steps to calculate IG(p2):



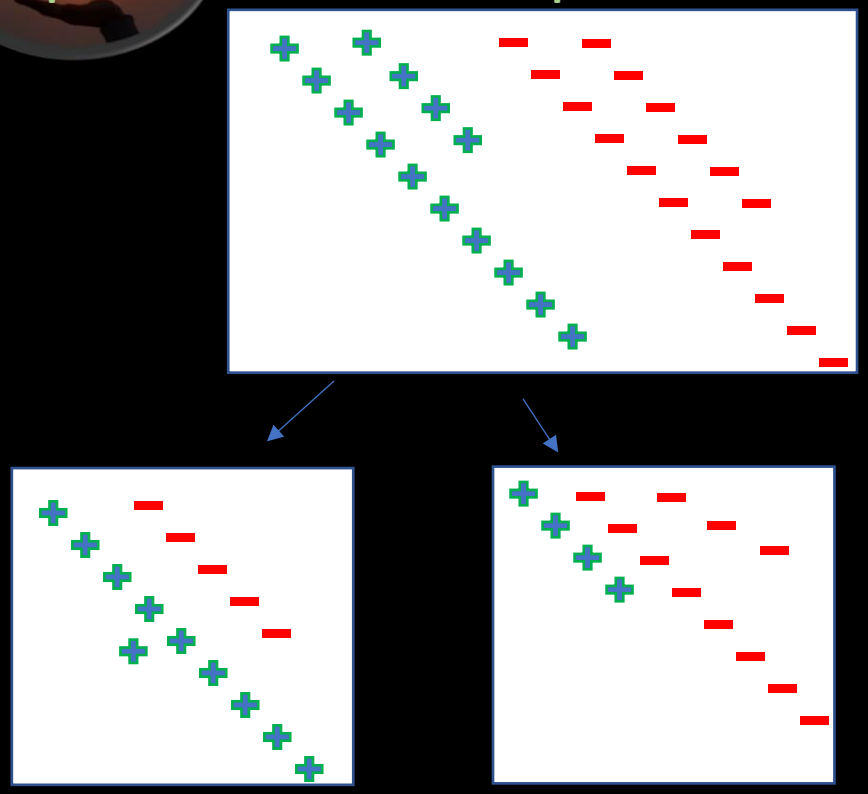
Step 1: Calculate Entropy of the Parent Node

positive cases(Has heart disease) = 14
negative cases(Has no heart disease) = 16
Total case = 30

Entropy of parent
= Entropy before split
= −[(14/30 * log(14/30) + 16/30*log(16/30)]
= −(0.466×−1.099+0.533×−0.912) = **0.999**
This is almost 1…so **high entropy.**

Steps to calculate IG(p3):

Step 2: Perform 3 Different Splits & Calculate Information Gain



Split 1: Based on Age Group (Younger vs. Older)

We split the data based on whether a person is younger (<40 years) or older (≥40 years).

Age Group	Positive Cases	Negative Cases	Total
Younger (<40)	10	5	15
Older (≥40)	4	11	15

Entropy of younger Node

= −[10/15 * log(10/15)+5/15*log(5/15)]

= −(0.666×−0.585+0.333×−1.585) = 0.918

Entropy of older Node

= −[4/15 * log(4/15)+11/15*log(11/15)]

= −(0.266×−1.907+0.733×−0.448) = 0.836

Entropy after split on Age group

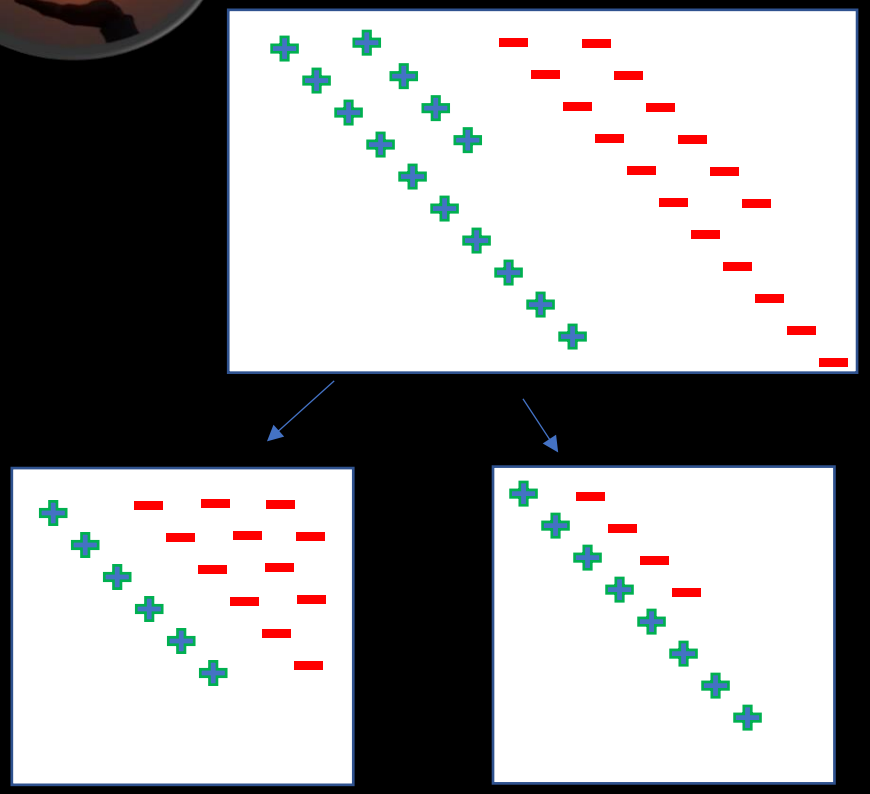
= (15/30×0.918)+(15/30×0.836) = 0.877

Info gain = Entropy before split - Entropy after split

= Entropy of parent - Entropy after split

= 0.999−0.877= **0.122**

Steps to calculate IG(p4):



Split 2: Based on Smoking Status (Smoker vs. Non-Smoker)
We split the data based on whether a person is smoker or not

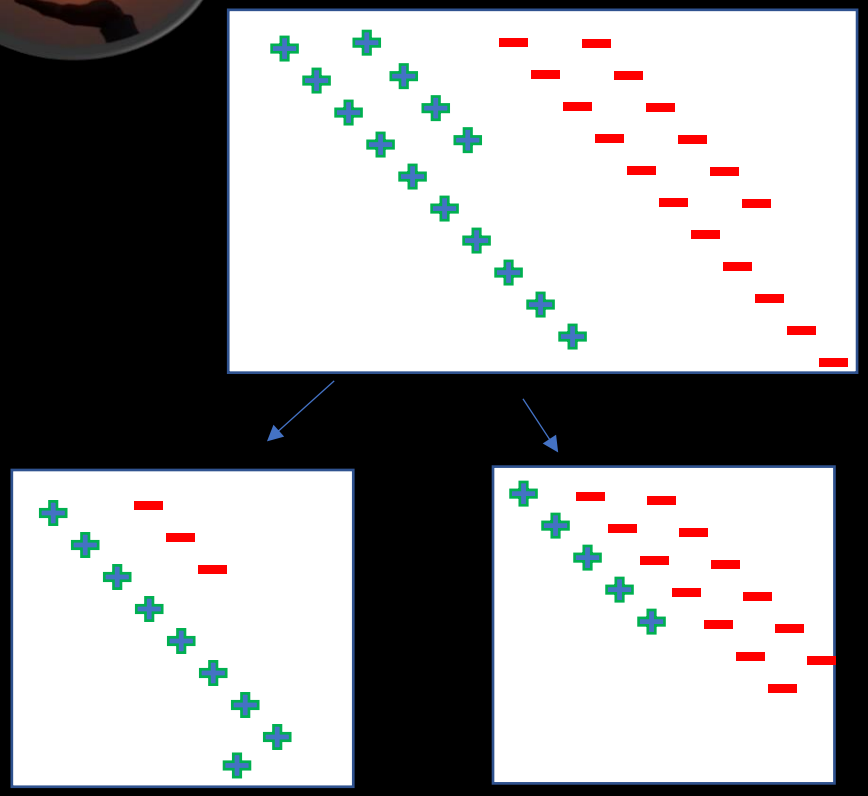
Smoking Status	Positive Cases	Negative Cases	Total
Smoker	6	12	18
Non-Smoker	8	4	12

Entropy of smoker Node
= −[6/18 * log(6/18)+12/18*log(12/18)]
= −(0.333×−1.585+0.666×−0.585) = 0.918
Entropy of non smoker Node
= −[8/12 * log(8/12)+4/12*log(4/12)] = 0.918

Entropy after split on smoking status
= (18/30 × 0.918) + (12/30 × 0.918)=0.918

Info gain = Entropy before split - Entropy after split
= Entropy of parent - Entropy after split
= 0.999 − 0.918 **= 0.081**

Steps to calculate IG(p5):



Split 3: Based on Exercise Habit (Regular vs. No Exercise)

We split the data based on exercise habit

Exercise Habit	Positive Cases	Negative Cases	Total
Regular	9	3	12
No Exercise	5	13	18

Entropy of regular-exercise Node
= −[9/12 * log(9/12)+3/12*log(3/12)] = 0.811

Entropy of no-exercise Node
= −[5/18 * log(5/18)+13/18*log(13/18)] = 0.850

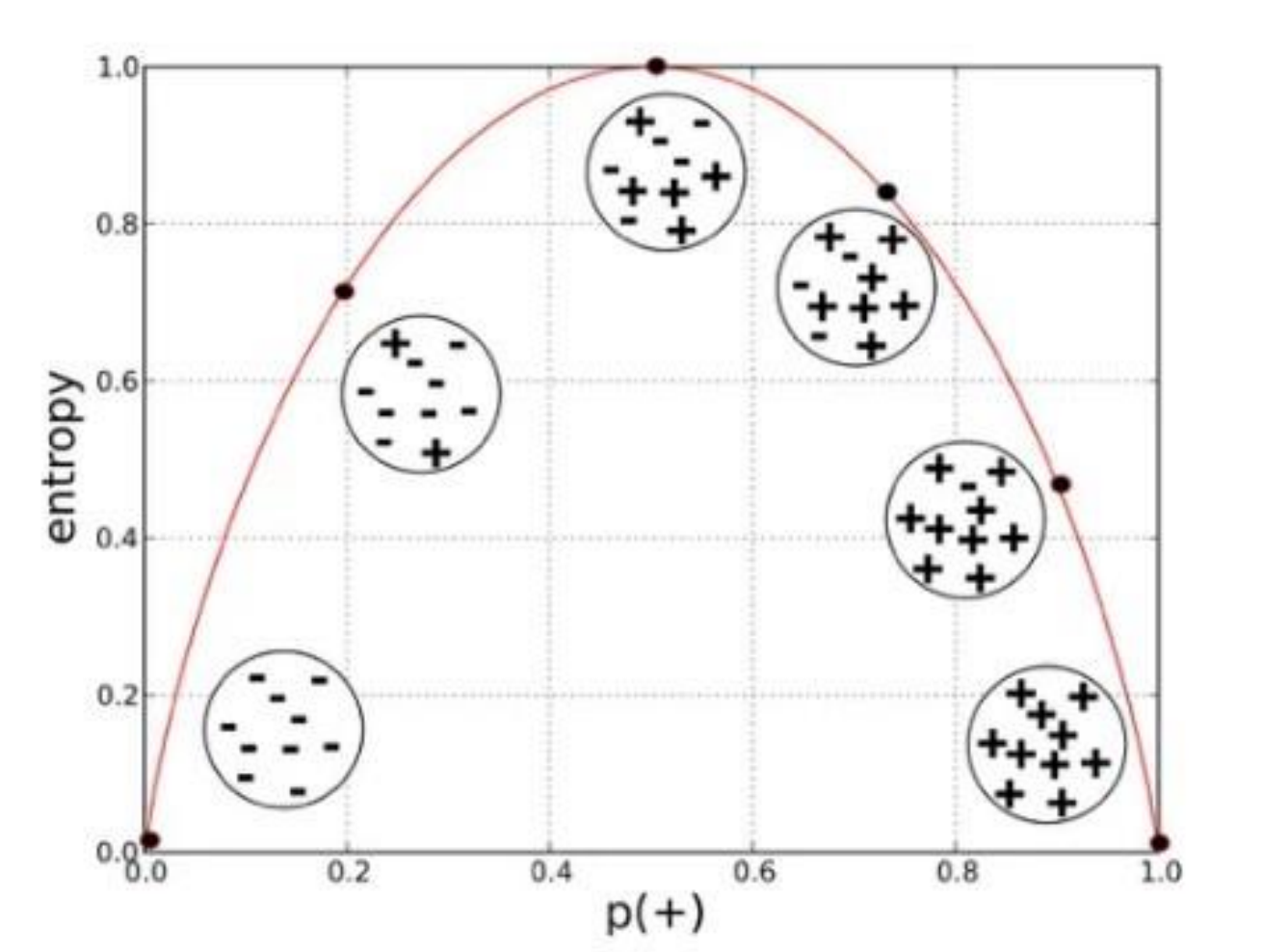
Entropy after split on exercise habit status
= (12/30×0.811) + (18/30×0.850) = 0.835

Info gain = Entropy before split - Entropy after split
= Entropy of parent - Entropy after split
= 0.999 − 0.835 = **0.164**

Split Criterion	Information Gain (IG)
Age Group (Young vs. Old)	0.122
Smoking Status (Smoker vs. Non-Smoker)	0.081
Exercise Habit (Regular vs. No Exercise)	0.164

Conclusion:
The best attribute for the split is **Exercise Habit (IG = 0.164)** as it gives the highest Information Gain.

How is entropy related to probability of positive class for **binary classification**?



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